

Example: Pattern matching, waves



[Mathphy](#)

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This works as an intuition building for understanding the Fourier transform.

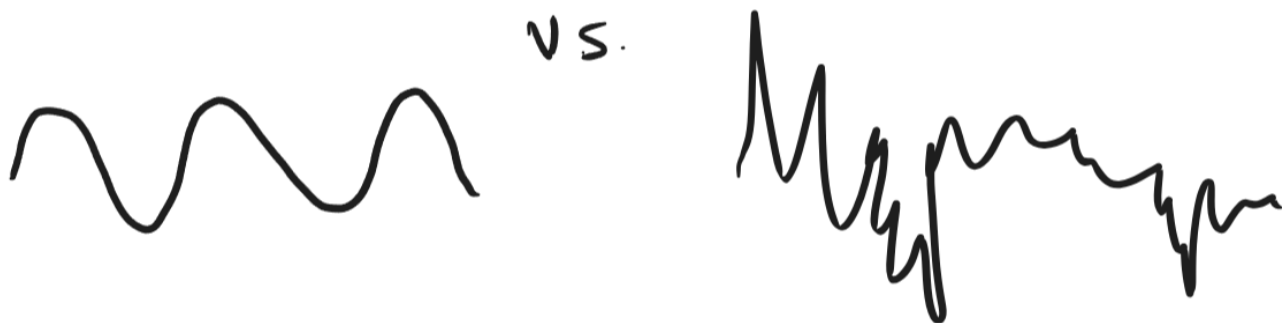
Recently, this other story told us an example of how we would find instruments that are in a mix of sounds:

Example: Pattern matching with sounds

There are 5 different instruments, and we mix 3 of them in 10 samples, our task is to discover which instruments are...

medium.com

In this other story we will be doing something similar, but instead of mixing instruments we will mix perfect sine waves. Note: musical instruments are a bandwidth of frequencies, so we can think of sine waves an atomic instrument, while complex instruments like a piano are a mix of waves.
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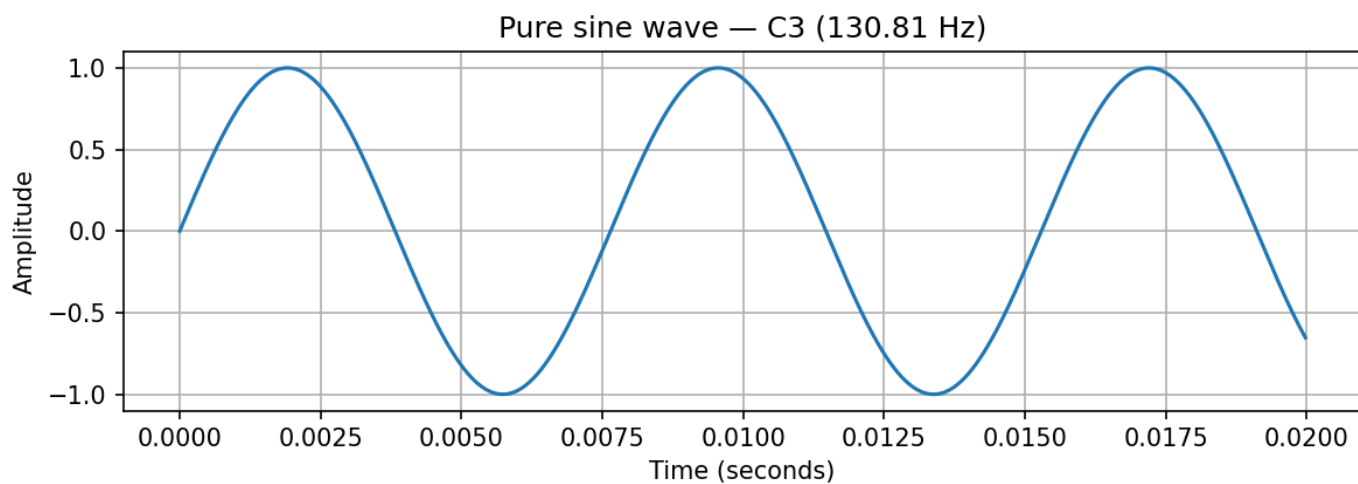
We will have in total 5 waves of different frequencies, we will have 10 mixes of them. Mind it as different frequencies are different instruments now:

Now we do the mix of 3 of them.

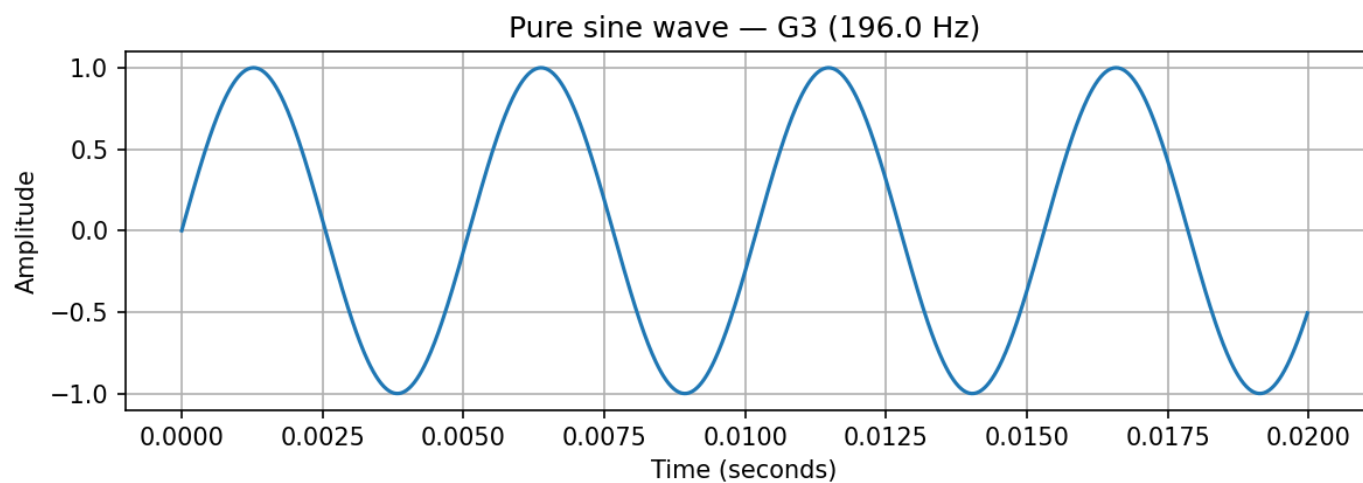
Again, we do the dot product between them:

Graphs

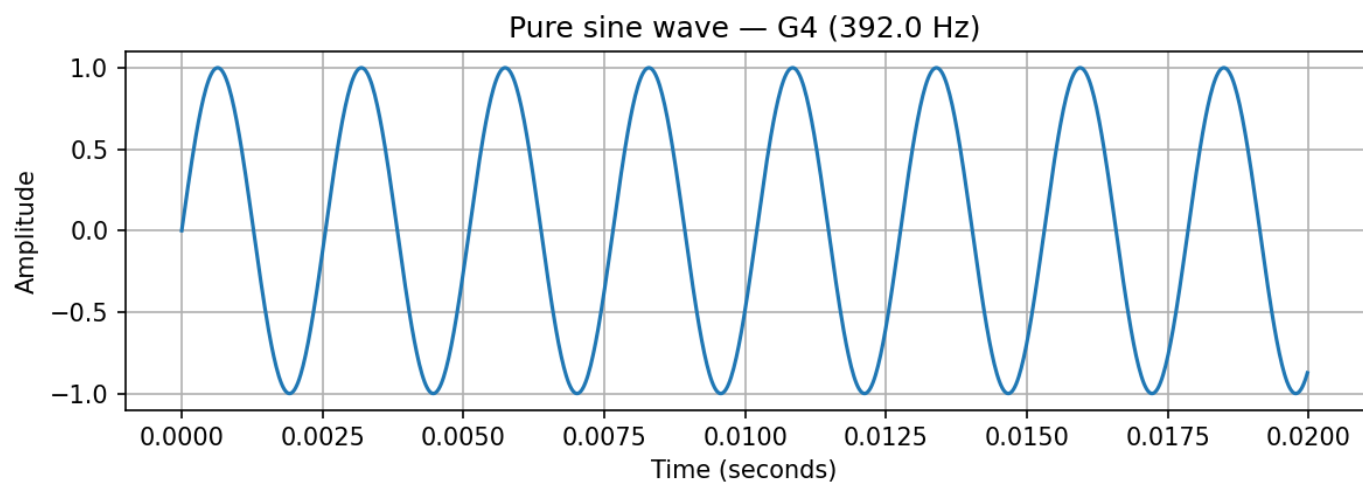
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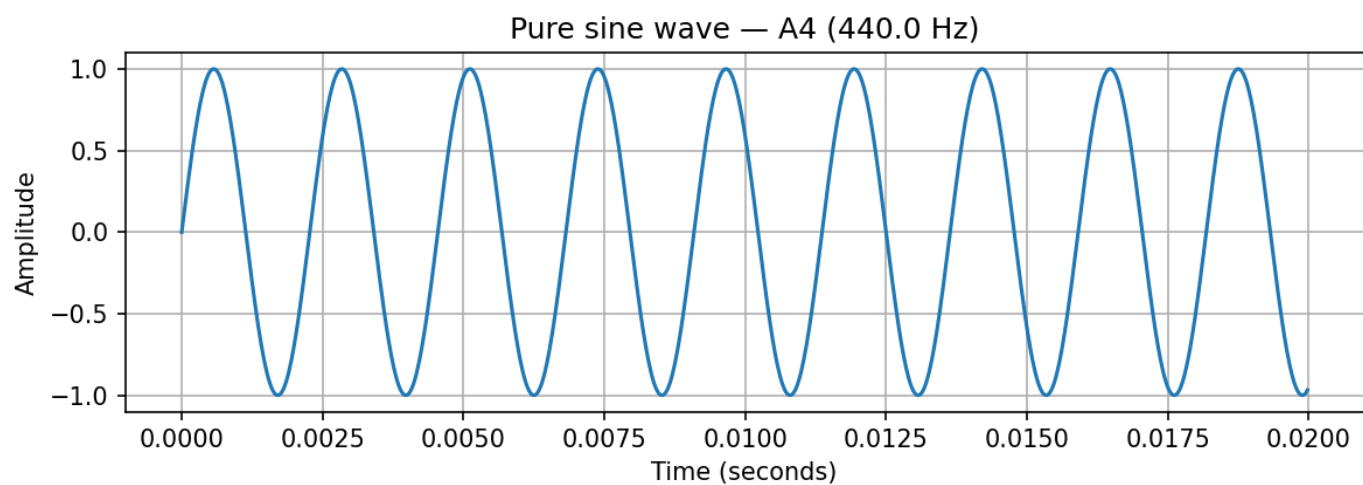
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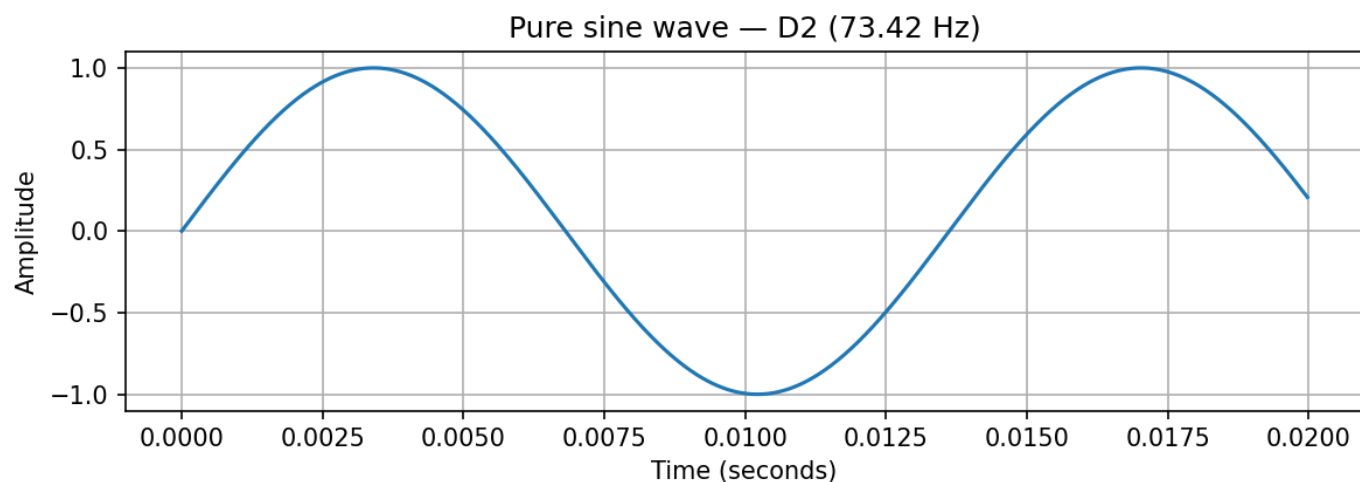
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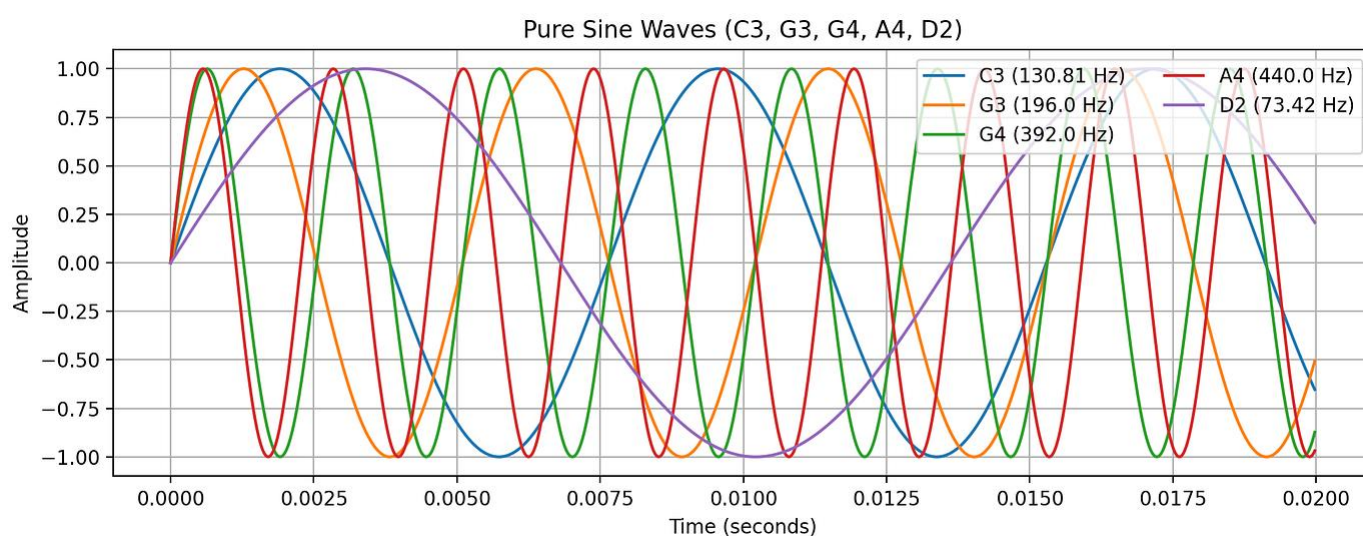


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All in one:

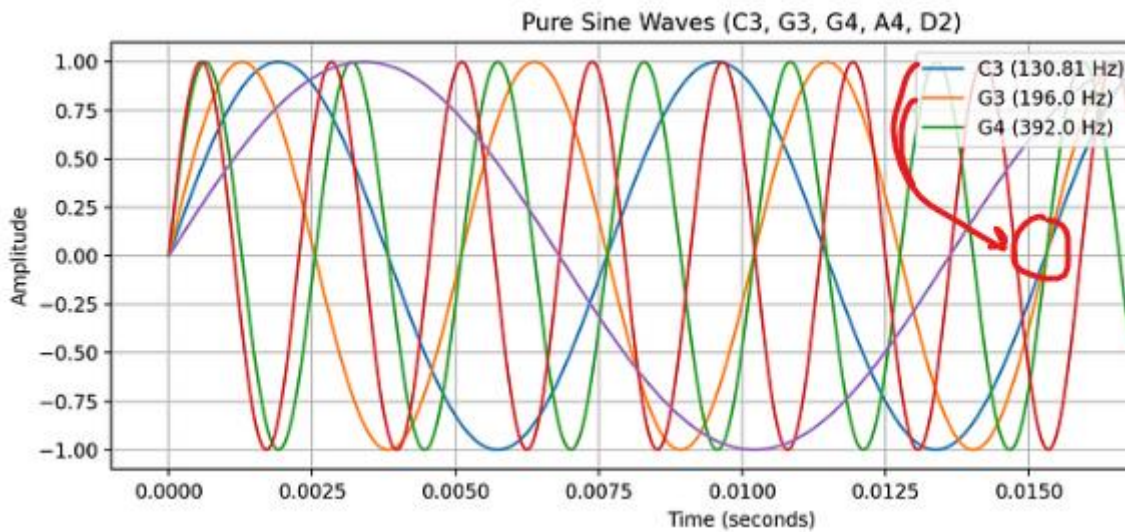
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Observations:

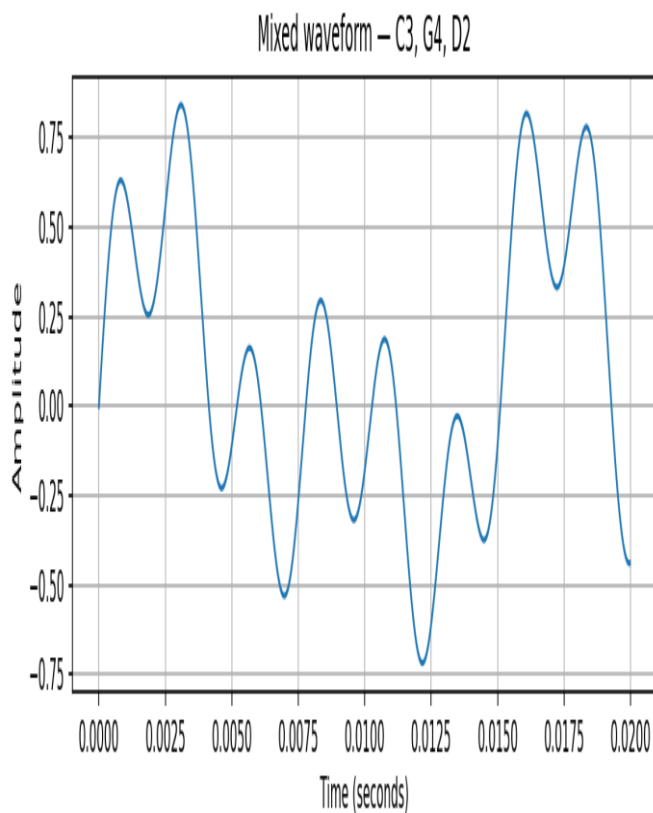
- First, the x-axis, as $A4 = 440\text{Hz}$ it means one oscillation each $\frac{1}{440} = 0.002272$.
- Second, the notes $G3$ and $G4$ known as octaves, are multiple of each other. $2 \times G3 = G4$, $2 \times 196\text{Hz} = 392\text{Hz}$. This means every 2 cycles of one, is 1 cycle of the other.

- Third, the notes **C3** and **G3** are perfect fifth, even if they are not multiple of each other, they are quite similar to be. In this fifth, they meet at 2 cycles:

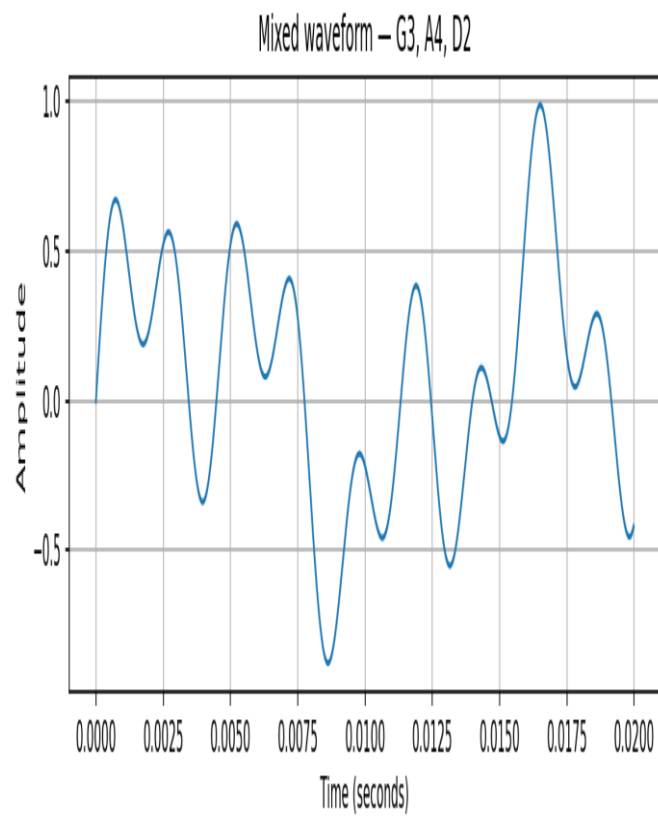


Now the graphs of the mixes:

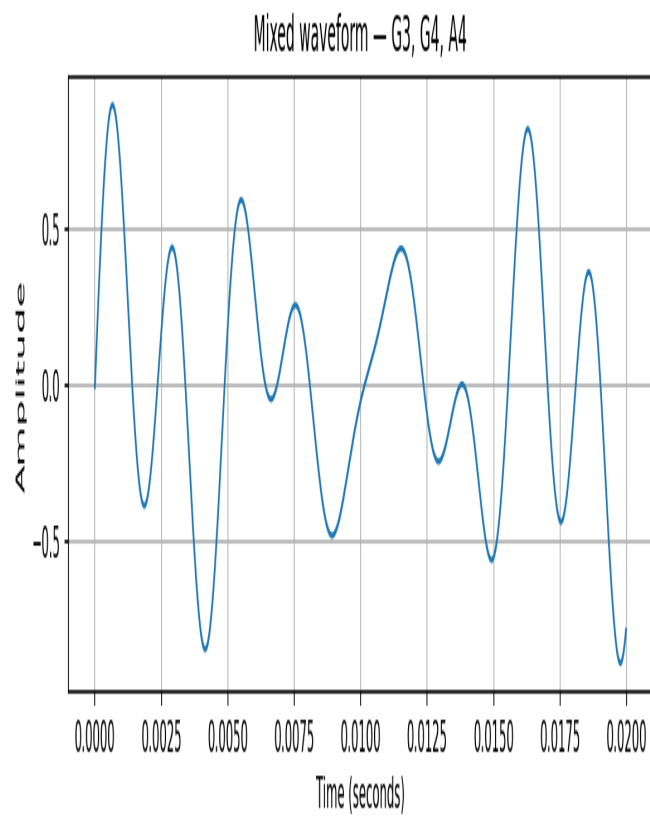
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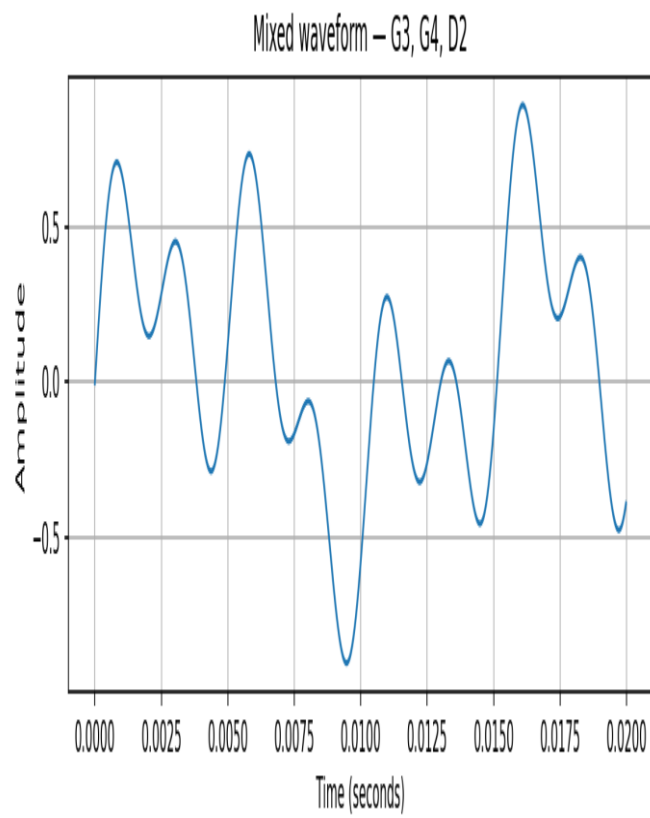
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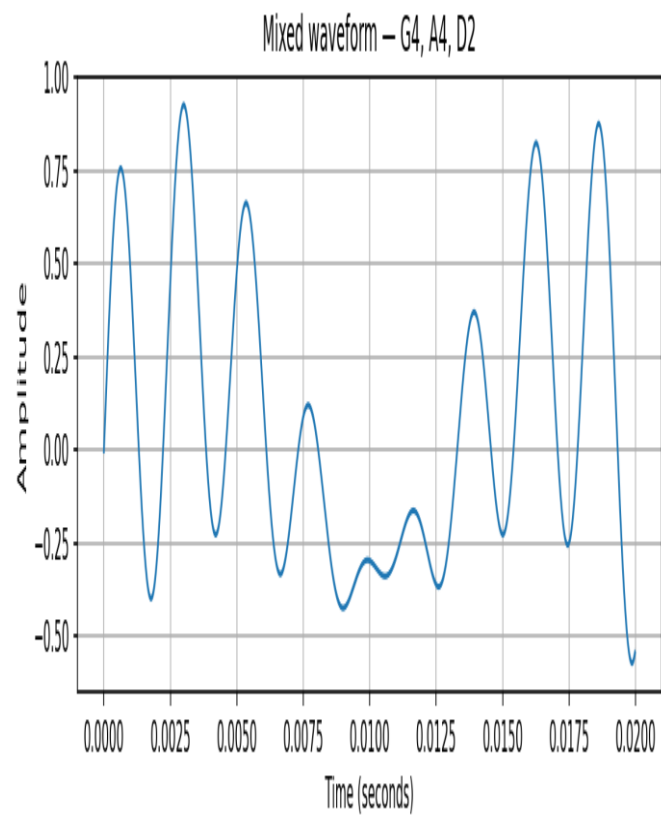
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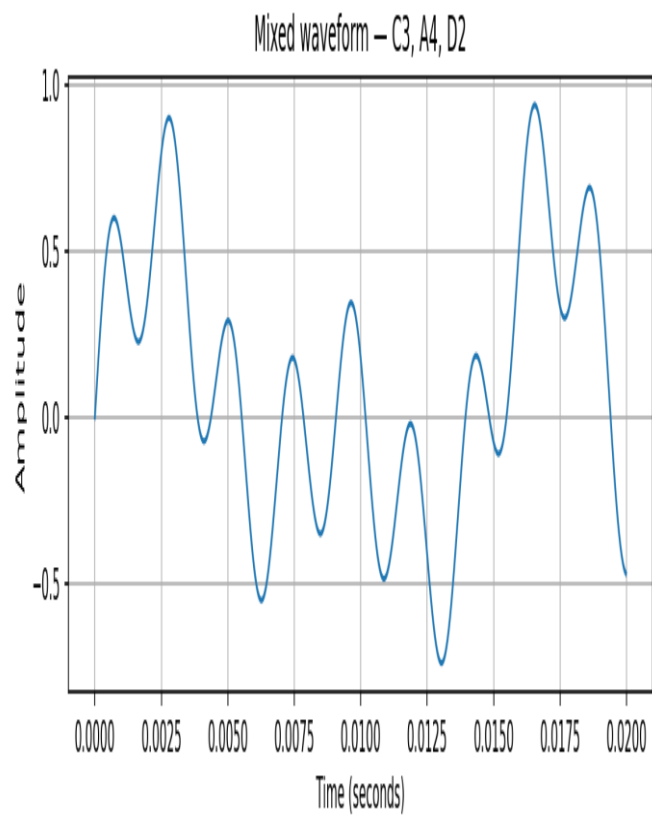
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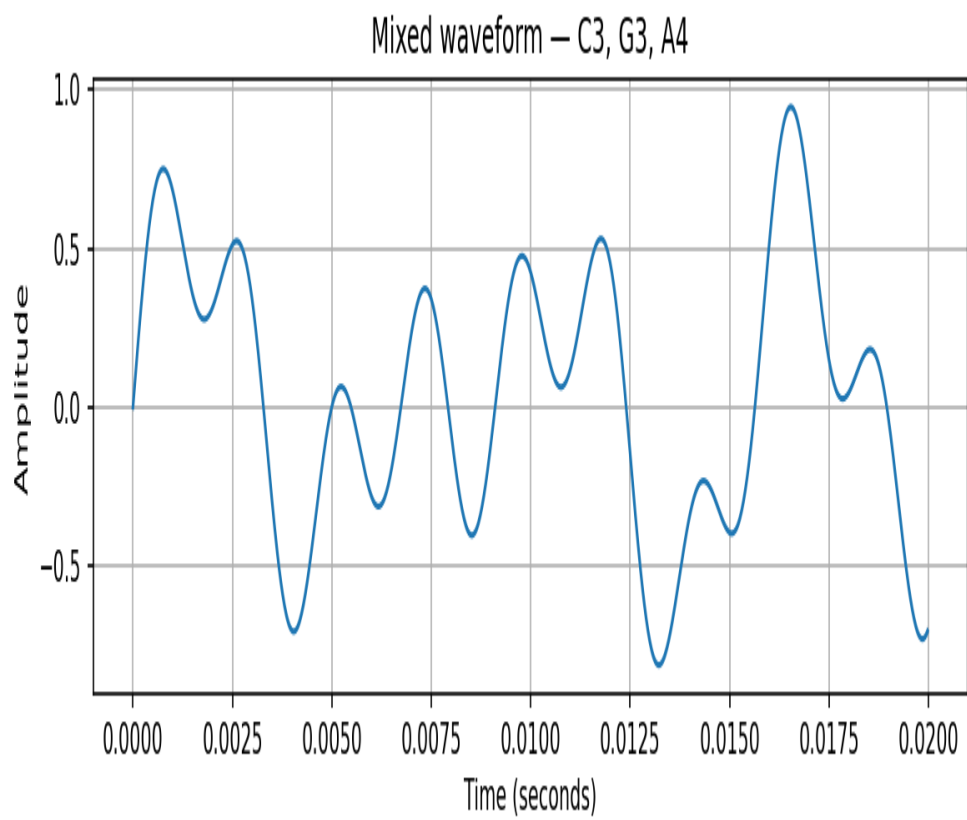
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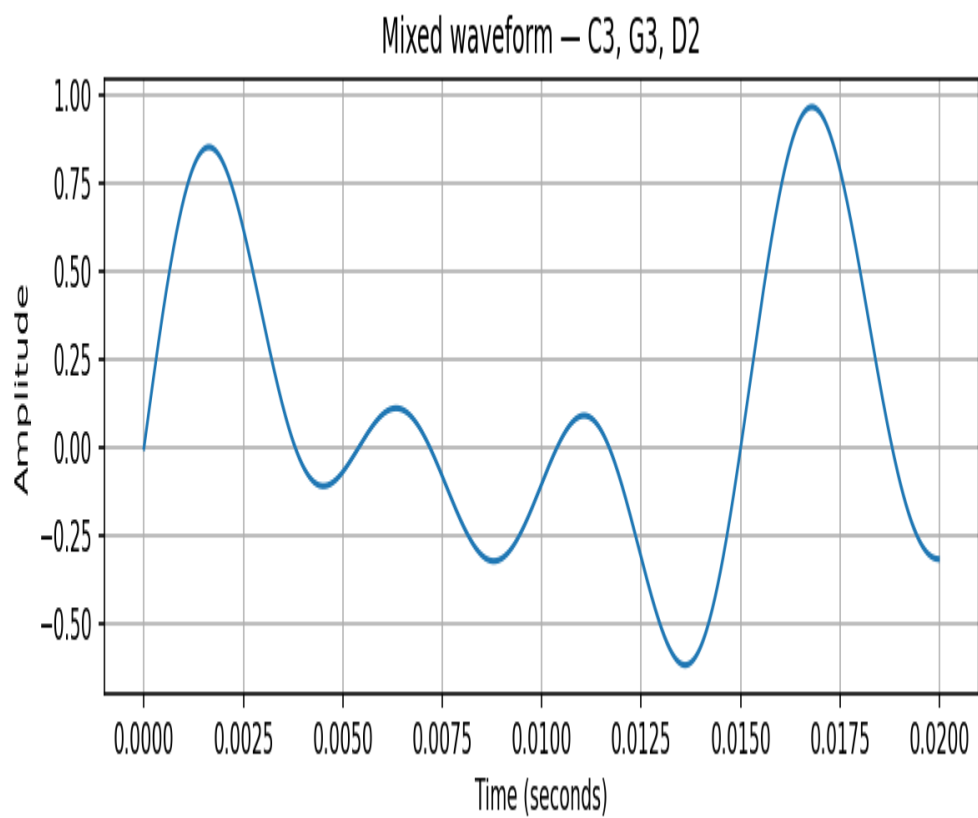
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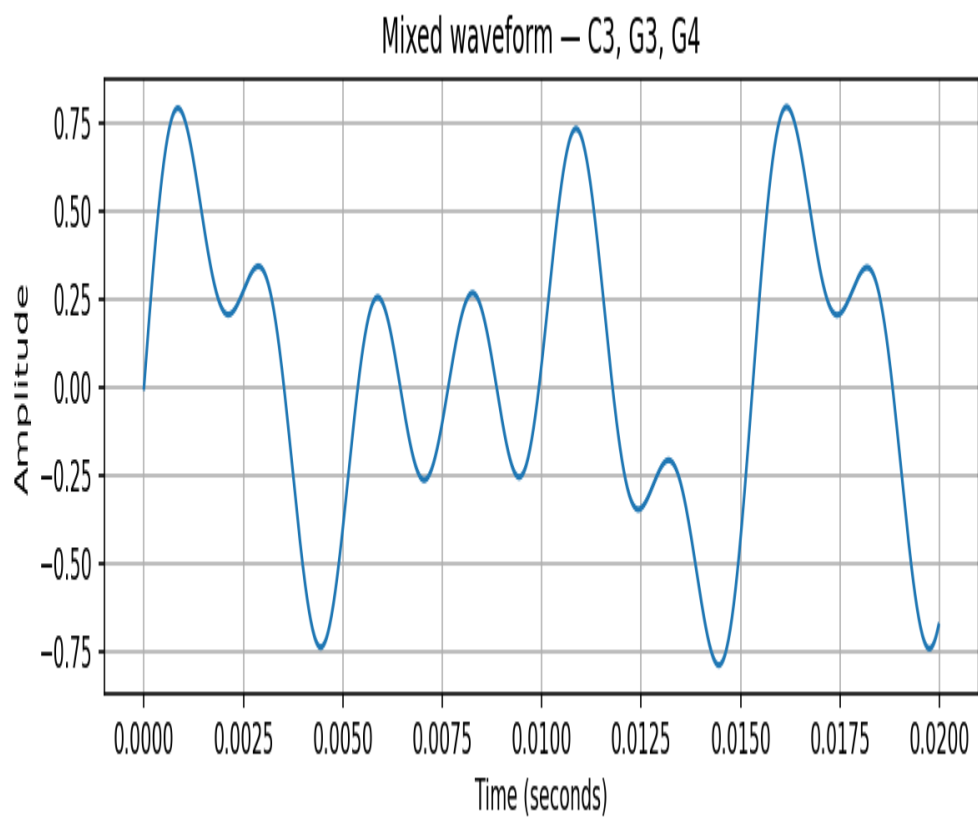
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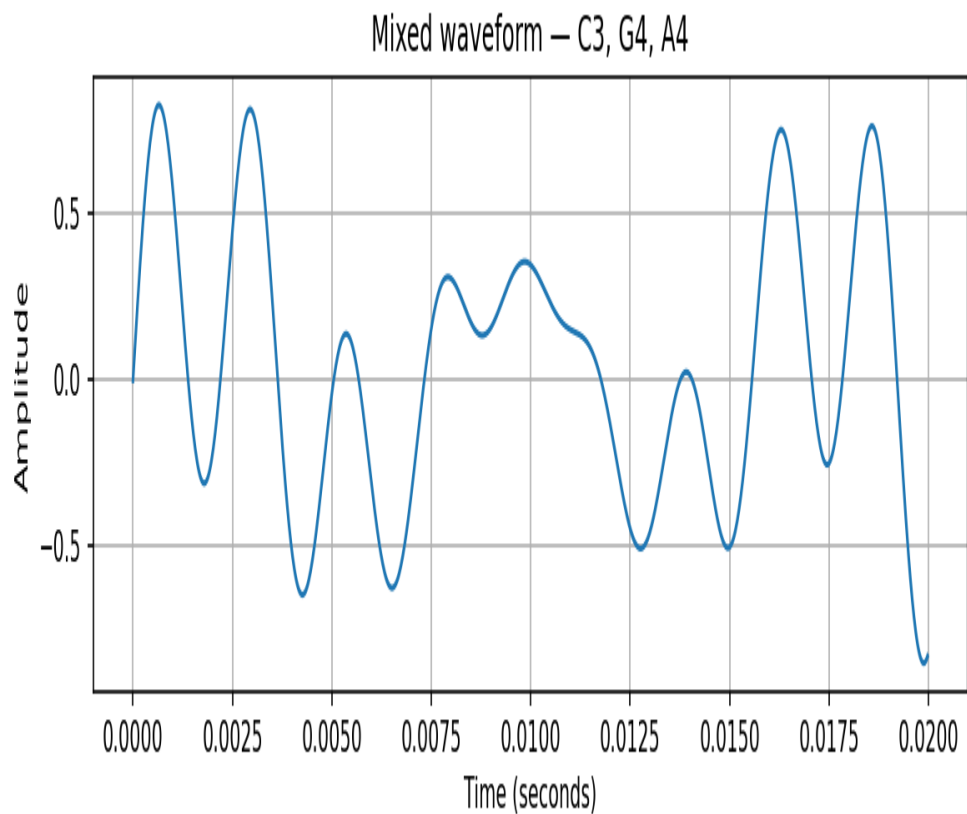
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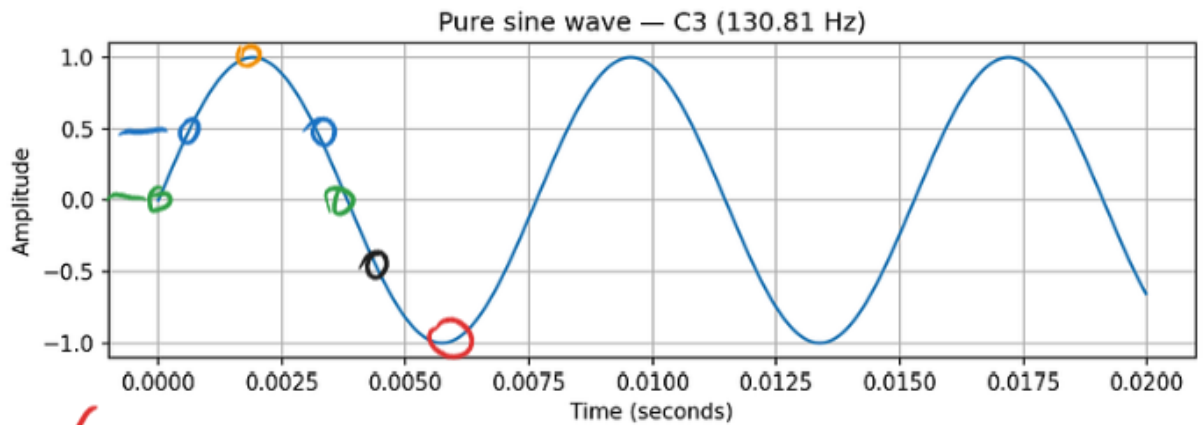
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The result of the normalized dot product:

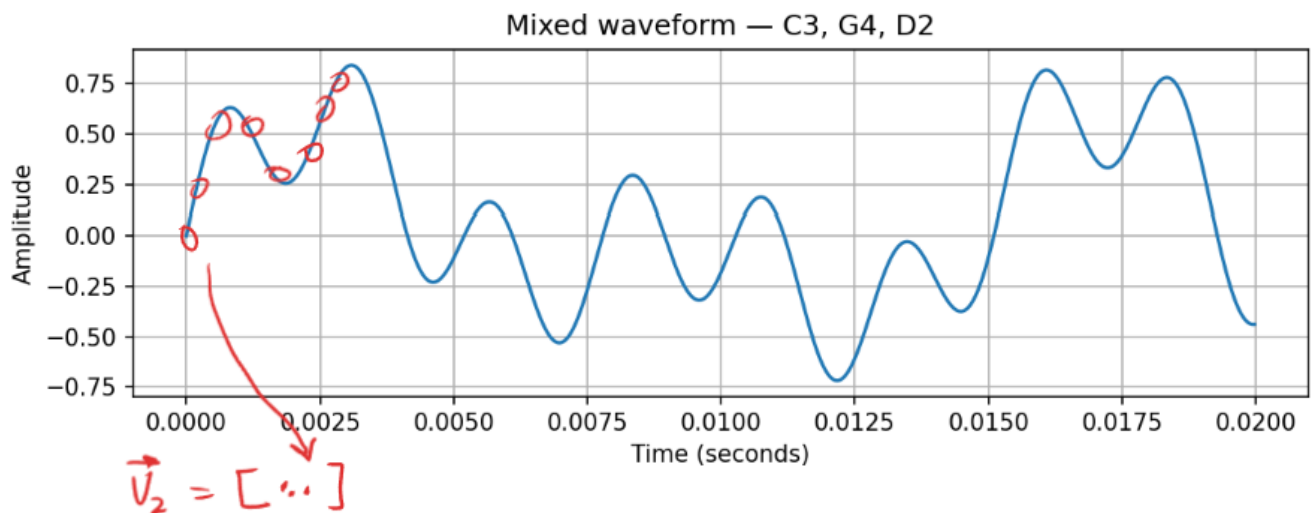
The normalized dot product is calculated as; take the sequence of an instrument (sine wave, in this example), multiply it in order, the first with the first, the second with the second, ... with the mixed sequence. Add it all and to normalize divide by the self-dot product.

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→ $\vec{v}_1 = [0, 0.5, 1, 0.5, 0, -0.5, -1, \dots]$

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The normalized dot product (cosine²) follows as:

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$$\text{Result} = \frac{(\vec{v}_1 \cdot \vec{v}_2)^2}{(\vec{v}_1 \cdot \vec{v}_1)(\vec{v}_2 \cdot \vec{v}_2)}$$

And that is what is shown here:

 analyzing mix_G4_A4_D2.wav

A4.wav: 0.3333

C3.wav: 0.0000

D2.wav: 0.3331

G3.wav: 0.0000

G4.wav: 0.3333

 analyzing mix_C3_A4_D2.wav

A4.wav: 0.3329

C3.wav: 0.3338

D2.wav: 0.3337

G3.wav: 0.0000

G4.wav: 0.0000

 analyzing mix_C3_G3_A4.wav

A4.wav: 0.3334

C3.wav: 0.3331

D2.wav: 0.0000

G3.wav: 0.3332

G4.wav: 0.0000

 analyzing mix_C3_G3_D2.wav

A4.wav: 0.0000

C3.wav: 0.3338

D2.wav: 0.3336

G3.wav: 0.3326

G4.wav: 0.0000

 analyzing mix_C3_G3_G4.wav

A4.wav: 0.0000

C3.wav: 0.3331

D2.wav: 0.0000

G3.wav: 0.3332

G4.wav: 0.3334

 analyzing mix_C3_G4_A4.wav

A4.wav: 0.3333

C3.wav: 0.3333

D2.wav: 0.0000

G3.wav: 0.0000

G4.wav: 0.3333

 analyzing mix_C3_G4_D2.wav

A4.wav: 0.0000

C3.wav: 0.3338

D2.wav: 0.3337

G3.wav: 0.0000

G4.wav: 0.3329

 analyzing mix_G3_A4_D2.wav

A4.wav: 0.3334

C3.wav: 0.0000

D2.wav: 0.3330

G3.wav: 0.3331

```
G4.wav: 0.0000
```

```
🎧 analyzing mix_G3_G4_A4.wav
```

```
A4.wav: 0.3333
```

```
C3.wav: 0.0000
```

```
D2.wav: 0.0000
```

```
G3.wav: 0.3333
```

```
G4.wav: 0.3333
```

```
🎧 analyzing mix_G3_G4_D2.wav
```

```
A4.wav: 0.0000
```

```
C3.wav: 0.0000
```

```
D2.wav: 0.3329
```

```
G3.wav: 0.3331
```

```
G4.wav: 0.3333
```

It perfectly matches the notes.

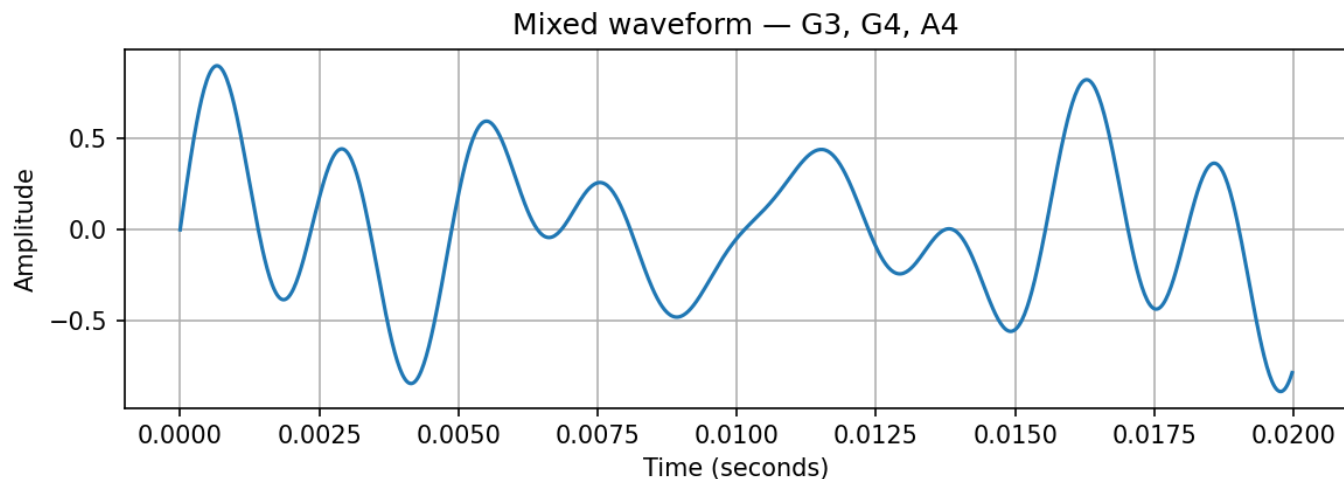
To think

why it didn't work so well in the example of instruments in "[Example: Pattern matching with sounds](#)" in there already are discussed some reasons, but to complement:

Think of instruments as a mix of waves, it still has additional differences, but the main thing that makes a musical instrument unique and so diverse (are tons of different instruments) is that they are combinations of sine waves, just as mixes in this story example. Also applies for human voice uniqueness.

You can appreciate how mixes start shaping complex waves that do not look exactly as a sine wave:

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Instruments are just like that, but instead of only 3 waves mixed, are many many, more than 1000 (or even infinite). The variety of instruments is the variety of combinations of the sine mix waves.

The dot product works well here because these waves are clean, they are independent from each other (the dot product of 2 waves is zero), so they do not interfere into finding the result. It would be difficult if each instrument had more that +1000 waves and some of them shared between different instruments.

Other reason why spectral analysis (Fourier) works for waves, is that it separates the volume from the note. The volume stop caring when we normalize, as the volumen or intensity of a wave is it's self-dot product.

Also, before if a high frequency instrument was mixed with a low frequency, this would lead to only appreciate the dominant one (low frequency), with Fourier this doesn't happen, it detects accurately different frequencies.

What has this to do with the Fourier transform?

This whole example is just a discrete Fourier transform, that is what it is: projecting functions (which means applying the dot product) into sine waves (the Fourier bases). We did this for mixes.

When we do spectral analysis (at using the Fourier transform) what we do is taking a mixed signal, decompose it into sine waves (match the instruments to find which instruments are in the mix, but this time the instruments are sine waves). Finally, by knowing which instruments (sine waves in this story) are in a mix, you can know more about the mix.

Why Fourier analysis has so many applications in applied physics? in summary because nature express itself in waves. So, it is easy to analyze it in its own language, just as it is written in nature. Just as it is for music and musical instruments too.

What is missing to complete this to be the actual Fourier transform? In this example we only used 3 sine waves (instruments), but while the discrete Fourier transform has thousand (depends of sample application), the **continuous** Fourier transform, known as only the “Fourier transform” has infinite sine waves as instruments. For example, frequencies from 0 to 10 in real numbers is an infinite set, but although it can be worked with finite ranges, it also works for infinite ones; from -infinity to infinity.

But the idea is the same, what the Fourier transform does is using sine waves as instruments, and finding the frequencies of any signal or function. It is just rewriting information using sine waves as its atoms.

Thanks for reading!

Mathematics

Math

Physics

